

The Long Reach of Targeted Smaller Classes: Evidence from a National Reform

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Abstract

This paper examines whether the effects of class size reduction in early primary grades persist through entry to middle school. I exploit France's national reform, launched in 2017, which targeted all grade 1 and grade 2 classes in disadvantaged schools, and produced positive and significant effects on French and mathematics achievement by the end of second grade. Using administrative data spanning eight cohorts of students entering sixth grade from 2017 to 2024, I employ an event study design comparing the evolution of sixth-grade outcomes between students exposed to smaller classes in early primary school and those who were not. I find a positive effect on mathematics achievement that emerges and strengthens across successive treated cohorts. The reform is also associated with an increase in students' self-reported academic confidence. These findings suggest that class size reduction, when targeted at the students expected to benefit most, produces effects that persist well beyond the early grades in which it was implemented.

Keywords: Class size, education policy, disadvantaged schools, persistence

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1 Introduction

The effects of class size on student performance have been the subject of extensive debate in both the scientific community and the political sphere (Hanushek, 2002; Blatchford & Russell, 2020; Charousset et al., 2025). The empirical evidence is mixed: while some studies find limited or absent effects (Hoxby, 2000; Blatchford et al., 2003; Bressoux et al., 2019), others document positive effects, both short-term and long-term (Krueger, 1999; Chetty et al., 2011; Fredriksson et al., 2013). A recurring finding across this heterogeneous evidence base is that, where positive effects are detected, they tend to be larger and more persistent for disadvantaged students (Krueger, 1999; Blatchford et al., 2003; Fredriksson et al., 2013). This pattern has largely been established through heterogeneity analyses conducted within broader, more socioeconomically mixed samples. A natural question is whether these findings are corroborated when a policy targets specifically the population for which class size reduction is expected to matter most, and whether any resulting gains persist over time.

France’s large-scale class size reduction reform, launched in 2017 and targeted exclusively at disadvantaged schools, offers an ideal setting to examine this question. These schools are classified since 2015 into two categories: Priority Education Networks (REP) and Reinforced Priority Education Networks (REP+).² The reform reduced grade 1 class sizes to approximately 12 students in REP+ schools (roughly half the pre-reform average) beginning in 2017-2018, then extended to grade 2 in REP+ and to grades 1 and 2 in REP schools in 2018 and 2019, respectively. In total, the measure led to the creation of 10,800 grade 1 and grade 2 classes in REP and REP+ schools, and an equal number of new teaching positions (Assemblée Nationale, 2023).

The DEPP (Directorate for Evaluation, Forecasting and Performance) evaluated the short-term effects of this policy using a representative sample of students from the first cohort of beneficiaries, who entered grade 1 in 2017. The evaluation revealed significant positive effects of 0.14 standard deviations in French and 0.12 standard deviations in mathematics at the end of grade 1, persisting through the end of grade 2 (Andreu et al.,

²At the start of the 2017 school year, priority education comprised 1,097 networks, including 365 middle schools and 2,463 elementary schools in REP+, as well as 732 middle schools and 4,237 elementary schools in REP.

2021). This study examines whether these effects persist at the beginning of secondary school (grade 6). I employ an event study design comparing sixth-grade achievement of students from REP+ schools, whose first treated cohort entered grade 6 in 2022-2023, with students from non-priority schools, never treated, and corroborate these estimates using a second comparison between students from REP schools, treated one year later, and students from non-priority schools. The analysis exploits data from national assessments administered to all sixth-grade students in France, covering eight cohorts entering grade 6 between 2017 and 2024.

The identification strategy relies on the parallel trends assumption: absent the reform, students from REP+ and non-priority primary schools would have experienced the same evolution in sixth-grade performance. I assess this assumption over the pre-reform period, noting one exception in 2020, the COVID-19 pandemic year, during which previous studies have documented differential impacts on students in disadvantaged schools (DEPP, 2022). I retain this year in the analysis but account for it explicitly throughout. Treatment assignment relies on students' grade 5 school sector classification, which may not perfectly reflect their school sector during grades 1 when they were treated. I document the extent of this measurement error using an exhaustive administrative panel tracking students from grade 1 through grade 5 (available from the cohort entering grade 1 in 2017 onward), and show that the main results are robust to two independent restrictions addressing these concerns directly.

The results reveal a positive and increasingly robust effect on mathematics achievement, and no detectable effect on French. In mathematics, an effect emerges for the second cohort in 2023 and grows further for the third cohort in 2024, reaching 0.07 standard deviations. This growing effect is robust to excluding grade repeaters, to restricting the sample to schools with stable student populations across cohorts, to using REP as an alternative treated group, and to a formal sensitivity analysis allowing for substantial deviations from parallel pre-trends. In French, no statistically significant effect is detected using either comparison group, and where parallel pre-trends hold most convincingly, the estimated effect remains null. The reform is also associated with a small but statistically significant increase in students' self-reported readiness to succeed, following the same delayed pattern observed for mathematics.

The estimated effect on mathematics achievement is smaller than the short-term effect of 0.12 standard deviations reported by Andreu et al. (2021) for the first treated cohort at the end of grade 2. Part of this difference may reflect measurement error arising from school mobility, which is likely to attenuate the estimates reported here toward zero. Despite this, the progressively larger effects on mathematics achievement observed for successive treated cohorts constitute the central finding of this study. These findings indicate that class size reduction is not only effective but also produces gains that persist over time when targeted at disadvantaged student populations.

1.1 Related Literature and Contribution

The effects of class size on student performance have been studied extensively, with mixed findings. The most influential evidence comes from the Student/Teacher Achievement Ratio (STAR) experiment, in which roughly 11,600 students and their teachers were randomly assigned to small classes (about 15 students), regular classes (about 22 students), or regular classes with an additional teacher, and followed from kindergarten through grade 3; Krueger (1999) finds that students in small classes scored 4 percentile points higher on standardized tests during their first year. This positive finding is corroborated by quasi-experimental designs exploiting class size thresholds: Piketty and Valdenaire (2006) exploit class opening thresholds to compare classes just above and below the threshold using a regression discontinuity design, and Urquiola (2006) applies a similar approach using a maximum class size rule, also finding a positive effect on student achievement. Yet other studies using similar designs find no effect: Hoxby (2000) exploits quasi-experimental variation in class size in the United States and finds no statistically significant effect on student performance, and Leuven, Oosterbeek, and Rønning (2008) exploit maximum class size rules in Norway and similarly find no effect on achievement. Among studies finding a positive effect on achievement, one natural question is whether these effects persist over the longer run. The evidence here is mixed and often points to fade-out on test scores specifically, even when effects on other outcomes persist: Krueger and Whitmore (2001) show that STAR's own test-score effects become statistically insignificant by grade 8, and Chetty et al. (2011) find that STAR participants nonetheless had higher college enrollment rates and higher earnings in adulthood, despite this fading test-score advantage. This study contributes to this question directly, building on a well-identified short-term

evaluation of the reform studied here (Andreu et al., 2021). I find that the short-term test-score gains it documents on mathematics persist and grow through secondary school entry, while the corresponding gains in French do not.

A second strand of this literature documents that class size reduction does not benefit all students equally, and tends to be more effective for certain populations: Krueger (1999) finds larger effects for students from low socioeconomic backgrounds in the STAR experiment, and Blatchford et al. (2003) show that effects are particularly marked for students with lower initial performance. This has shifted part of the policy debate toward whether class size reduction should be targeted specifically at the students expected to benefit most. This study speaks directly to this question: because the French reform targets disadvantaged schools exclusively, it allows for a direct test of whether class size reduction is effective when targeted at the population for which it is expected to matter most. I find that it is: the mathematics effect not only persists in this population but grows over successive cohorts, in contrast to the fade-out typically documented in less targeted settings.

Finally, most of this literature focuses on cognitive outcomes with little causal evidence on non-cognitive outcomes specifically. Dee and West (2011) re-analyze follow-up data from the STAR experiment and find that class size reduction in the early grades improved student initiative. However, this effect did not persist into grade 8, fading in a pattern similar to the test-score effects discussed above. This study contributes to this literature by examining whether exposure to smaller classes in grades 1 and 2 affects a non-cognitive outcome, students' self-reported readiness to succeed, five years after treatment. I find a small but persistent positive effect on this outcome, following the same delayed pattern as the mathematics results. This finding speaks directly to the puzzle described above: it is consistent with the possibility that some of the longer-run benefits of class size reduction documented elsewhere in the literature operate through non-cognitive channels not fully captured by test scores.

The paper proceeds as follows: Section 2 describes the institutional context, Section 3 presents the data, Section 4 outlines the empirical strategy, Section 5 reports the results, and Section 6 concludes.

2 Institutional Setting

France's education system is organized into three stages: primary school (*école primaire*), middle school (*collège*), and high school (*lycée*). Primary school spans five years, from grade 1 (*cours préparatoire*, or CP) through grade 5 (*cours moyen 2*, or CM2), typically for children aged six to eleven; students then transition to middle school (*collège*) at grade 6 (*sixième*), the entry point examined in this study. The system is highly centralized: the Ministry of National Education sets a single national curriculum, and all students take the same national assessments at grade 6 entry regardless of school or location, described further in Section 3. Public school assignment is determined by residential catchment areas (*carte scolaire*), so that a school's student population largely reflects the socioeconomic composition of its surrounding neighborhood.³ Because catchment-based assignment concentrates disadvantaged students in specific schools, France has since 1981 designated a subset of schools as priority education networks, granting them additional resources to address this inequality.

2.1 Priority Education

Priority education networks were created in 1981 with the objective of reducing educational inequalities by allocating additional resources to schools located in geographic areas with high concentrations of disadvantaged populations. The main objective of this policy is "to obtain a significant improvement in the academic results of students, particularly the most disadvantaged" (Circular n°90-028 of 01/02/1990). Initially conceived as a temporary measure, this policy has undergone several evolutions, expanding its initial scope (2.1).

2.1.1 Evolution of the Program

In 1990, a first reform revised the map of priority education zones (ZEP), increasing their number from 363 to 558. In 1999, the policy evolved with the creation of priority education networks (REP), which replaced ZEP by establishing network-based collaboration. The objective was for these networks, composed of schools and middle schools, to "pool their

³This is not always the case in socioeconomically mixed neighborhoods; see the IPP report on social mixing in Paris middle schools: <https://www.ipp.eu/rapport-note-ipp-renforcer-mixite-sociale-college-evaluation-secteurs-multi-colleges-paris/>.

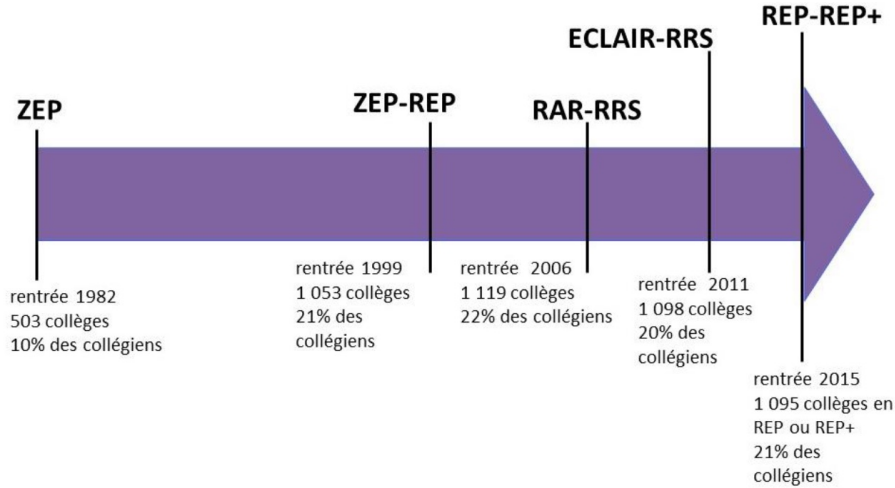


Figure 2.1: Evolution of Priority Education Programs in France

pedagogical and educational resources as well as their innovations in service of student academic success" (Circular n°99-007 of 20/01/1999). This reform redefined the map: more than 700 establishments exited, particularly rural schools, while 1,559 elementary schools, middle schools, or high schools entered. Since then, approximately one-fifth of public middle school students have been enrolled in priority education (Stefanou, 2022). The 2006 reform created "Ambition Success" Networks (RAR) and School Success Networks (RRS), structured around the middle school as the reference unit (Circular n°2006-058 of 30/03/2006). In 2010-2011, the ÉCLAIR program (Schools, Middle Schools, High Schools for Ambition, Innovation and Success) was piloted and extended to almost all RAR. Between 2014 and 2015, the priority education map was revised with a new organization into two types of networks: priority education networks (REP) and reinforced priority education networks (REP+). This reform is based on a social index incorporating four criteria: the proportion of students from disadvantaged backgrounds, scholarship recipients, students living in priority neighborhoods, and students with academic delay at sixth-grade entry. The middle school remains at the center of the network, and elementary schools are classified as REP or REP+ based on their assigned catchment middle school (collège de secteur). Resources are more concentrated on REP+, which group the most disadvantaged schools. High schools exited the program during this reform (Stefanou, 2022).

2.1.2 REP and REP+

REP+ targets "neighborhoods or isolated sectors experiencing the greatest concentrations of social difficulties having strong impacts on academic success," while REP includes schools "more socially mixed but encountering more significant social difficulties than those of middle schools and schools located outside priority education" (Ministry of Education). As of 2023, priority education comprises 1,093 networks, including 731 middle schools and 4,136 elementary schools in REP, and 362 middle schools and 2,458 elementary schools in REP+, serving more than 1.7 million students (DEPP, 2024).

2.1.3 Resource Allocation

Priority education provides differentiated resources to designated schools. Teachers receive financial bonuses, priority for transfers, and additional professional development time. In REP+, middle school teachers benefit from reduced teaching loads (1.1 weighting factor per teaching hour), and elementary teachers receive 18 half-days annually for collaborative work, individualized student support, and family outreach (Stefanou, 2022). The program also supports targeted interventions including homework assistance and mentoring programs. The reduction in class size represents a key policy lever, increasing the time of individualized instruction to improve learning conditions.

2.1.4 Class Size Reduction, a Progressive Measure Launched in 2017

In 2017, France implemented a class size reduction policy in first grade REP+ schools, targeting a maximum of 12 students per class—approximately half the pre-reform average. The program expanded to second grade REP+ in 2018, then to REP schools (first and second grade) in 2018-2019, and finally to kindergarten in both networks starting in 2020 (2.2). Schools could implement the reform through different modalities: opening new classes or co-teaching arrangements. Space constraints, particularly in large metropolitan areas and overseas departments, sometimes resulted in two classes sharing a single room (IGÉSR, 2023). Administrative data do not distinguish between these implementation modes. This study therefore estimates the average effect of the class size reduction policy across all modalities.

	2017	2018	2019	2020
Grade 1 REP+	First treated cohort	Treated	Treated	Treated
Grade 1 REP		First treated cohort	Treated	Treated
Grade 2 REP+		First treated cohort	Treated	Treated
Grade 2 REP			First treated cohort	Treated
Kindergarten (REP & REP+)				First treated cohort

Figure 2.2: Implementation Timeline

2.3 shows that the reform reduced average class size in treated grades from approximately 22 to 13 students between 2015 and 2020. The program required 15,987 additional full-time equivalent teaching positions, at an annual cost of approximately €800 million (Cour des comptes, 2025).

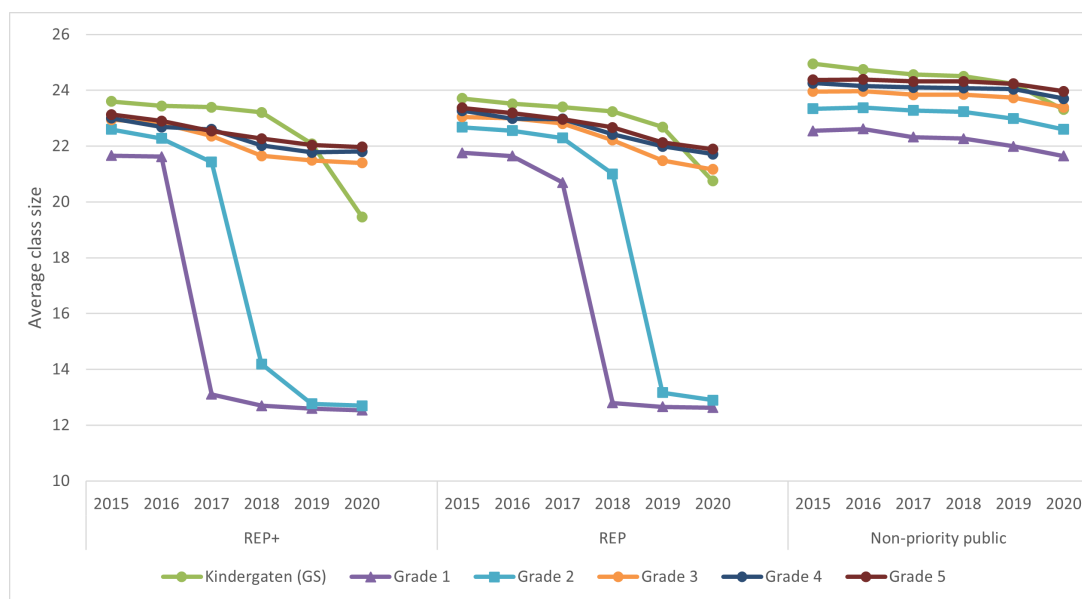


Figure 2.3: Evolution of Class Size, 2015-2020

DEPP evaluated the short-term effects of the policy using a sample of 15,000 students across 204 REP+ schools, 104 REP schools with similar characteristics ("close to REP+"), and 102 non-priority schools (Andreu et al., 2021). Students took standardized assessments at the beginning and end of first and second grade. The evaluation compared REP+ students (treated in 2017) with matched students from "close to REP+" schools (not yet treated), using propensity score matching based on school characteristics including the

proportion of students from disadvantaged backgrounds, grade retention rates, and median municipal income. The evaluation found significant positive effects at the end of first grade: students in reduced-size classes scored 0.14 standard deviations higher in French and 0.12 standard deviations higher in mathematics relative to the comparison group. These gains persisted through the end of second grade. Andreu et al. (2021) attribute the effects primarily to improved learning conditions—including more individualized teacher-student interactions and higher student engagement—rather than to changes in pedagogical practices, which remained largely unchanged. The study emphasizes that implementation quality, including local leadership and teacher support, played an important role in program effectiveness.

3 Data

3.1 Test Score Data

The data come from national assessments in French and mathematics administered to all sixth-grade students in public and private contract middle schools in France since 2017. These assessments, designed by DEPP in collaboration with the National Scientific Council of Education (CSEN) and the Directorate General of School Education (DGESCO), provide benchmarks to help teachers identify and respond to individual student needs.

I use sixth-grade assessments from 2017 to 2024. The dataset includes students' sex, date of birth, socioeconomic status, and information on both their sixth-grade middle school and fifth-grade elementary school, including academy, department, sector, and priority education status (private, public, REP, or REP+). Figure 3.1 presents the evolution of key indicators over this period.

The analysis uses students from a balanced panel of fifth-grade schools observed in every year from 2017 to 2024. The primary outcome variables are French and mathematics scores, standardized annually by subject (mean zero, standard deviation one). The main analysis includes 23,207 schools and 4.5 million students across the eight cohorts examined. Sample sizes for additional analyses are reported alongside each specification.

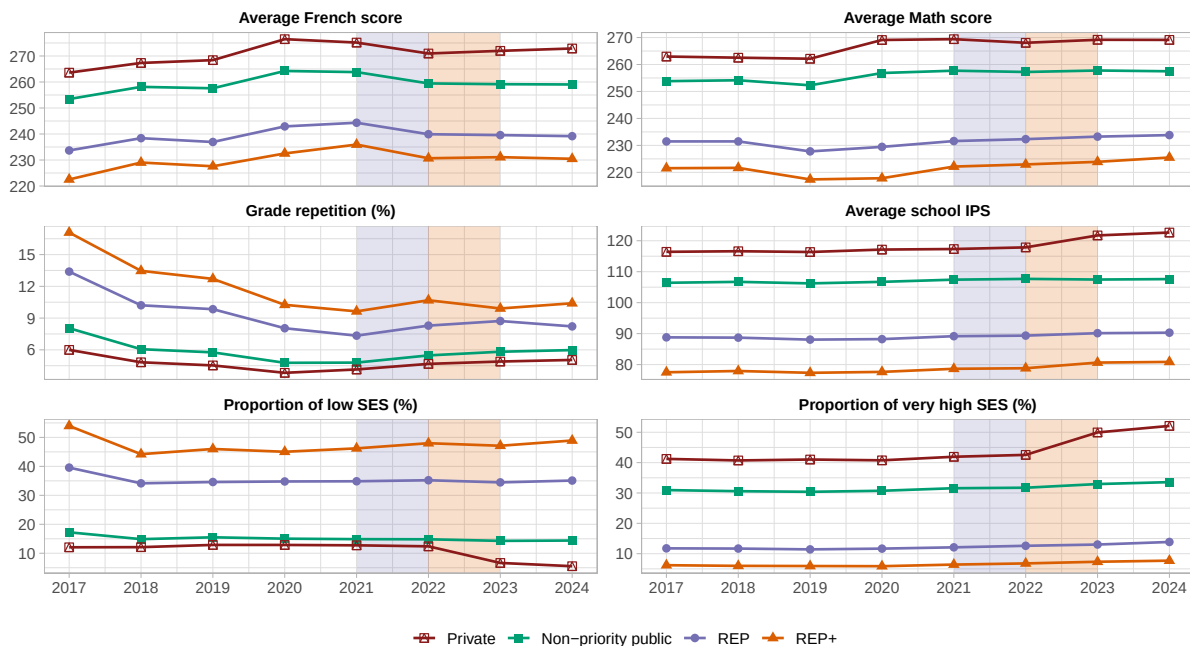


Figure 3.1: Evolution of Key Indicators at Sixth-Grade Entry, 2017-2024

Source: DEPP, National assessments at sixth-grade entry, 2017-2024. Sample includes all public and private schools in France. Shaded regions indicate reform implementation: purple (2021-2022) marks the transition to the first REP+ cohort treated in first and second grade; orange (2022-2023) marks the same transition for REP schools.

3.2 Socio-Cognitive Skills

Since 2020, sixth-grade assessments have included questions measuring students' socio-cognitive skills, capturing perceptions of self-confidence and sense of success beyond academic performance. These socio-cognitive skills play an important role in educational success: they are associated with better academic perseverance, greater motivation, and better learning trajectories (Heckman & Kautz, 2012; Borghans et al., 2008). Furthermore, recent research emphasizes that these dimensions significantly predict academic results and classroom behaviors, particularly among students from disadvantaged backgrounds (Cunha et al., 2010).

In 2020, one question was added: "At the beginning of this sixth-grade year, do you think you are ready to succeed in your year?" Students responded on a four-point Likert scale from "Not at all" to "Yes, absolutely." In 2021, two additional questions asked students to evaluate their perceived performance on the French and mathematics assessments using

the same scale.⁴ I use the 2020 question, described further in Section ??, to evaluate the reform’s impact on non-cognitive skills.

3.3 Treatment Assignment and Measurement Error

Sixth-grade assessments record the elementary school students attended in fifth grade. Figure 3.2 uses this variable to show the sector students are coming from at grade 6 entry. REP+ and REP schools show the lowest persistence of any sector, with roughly a quarter of students originating from these schools moving elsewhere by sixth grade, compared to under 16% for non-priority public and private schools. Among REP+ and REP movers, the most common destination is non-priority public schools (11.7% and 15.5% of students, respectively).

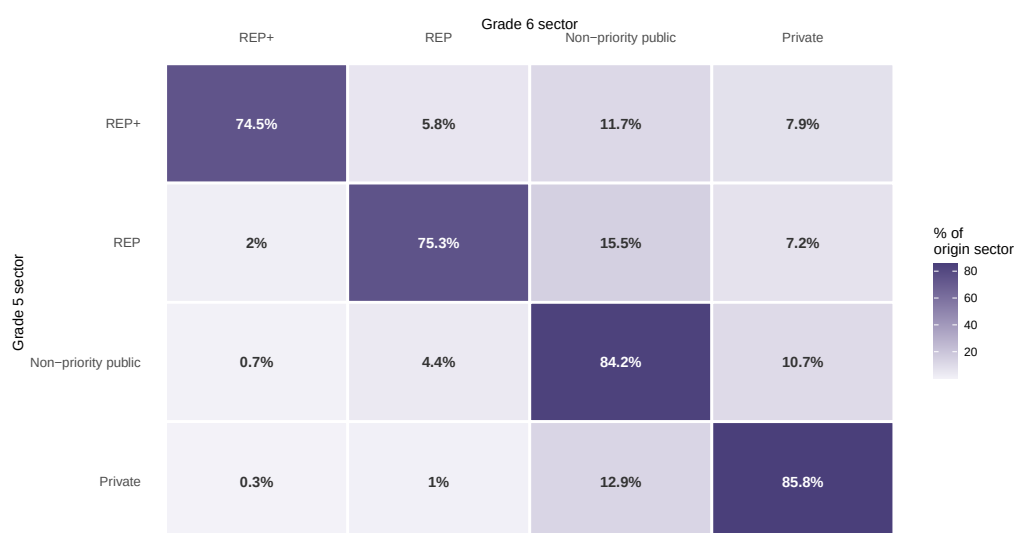


Figure 3.2: Sector Persistence and Mobility from Grade 5 to Grade 6

Source: DEPP, National assessments at sixth-grade entry, 2017-2024. Each row sums to 100% and shows the destination sector distribution of students originating in the row’s fifth-grade sector.

Since sixth-grade assessments do not record students’ first-grade schools, treatment and comparison groups are defined based on students’ fifth-grade school sector, which may not perfectly reflect their actual exposure to treatment during grade 1: a student’s fifth-grade sector can differ from their sector during the treated grades if they moved schools, or if grade retention shifted them into a different cohort than the one their treatment timing

⁴These questions appear at the end of assessments, so response rates may be slightly lower than for the main tests.

would imply. To document the extent of this mobility directly, I turn to a separate administrative source that tracks individual students across grades.

Using DIAPRE, an exhaustive administrative panel of all students entering public and private-under-contract elementary schools in France and available at the student level from 2017 onward, I construct a cohort tracking all students who entered grade 1 in 2017 through fifth grade in 2021. This allows me to observe each student’s actual school and sector between grade 1 and grade 5. I document three sources of measurement error in treatment assignment. First, 6.4% of REP+ students repeat at least one grade between grade 1 and fifth grade, comparable to the 6.7% repetition rate observed in REP and modestly higher than the 5.0-5.1% observed in non-priority schools; repetition may cause students to be tracked in a different cohort than their actual treatment timing implies. Second, among students who attended a REP+ school in grade 1, 14.0% were enrolled in a non-priority public school by fifth grade. These students would appear in the comparison group despite having received treatment. Third, among students classified as REP+ at fifth grade, 12.8% did not attend a REP+ school during grades 1. These students dilute the treatment group with untreated students. I repeat this analysis for students entering grade 1 in 2018 and 2019, tracked through the same panel construction. All three shares are stable across the three cohorts examined, indicating this reflects a structural feature of priority-education student mobility rather than a trend specific to any one cohort. Each of these misclassification channels is expected to bias estimates toward zero: repeaters and students who move into REP+ after the treated grades dilute the treatment group with untreated students, while students who move out of REP+ dilute the comparison group with treated students. As a result, the estimates reported in the Results section should be interpreted as intent-to-treat effects and likely represent a lower bound on the true effect of experiencing reduced class sizes during grade 1.

As robustness checks reported in Section A.4, I estimate the main specification on two restricted samples designed to address these mobility and grade repetition concerns directly. First, I restrict the analysis to on-time students, excluding grade repeaters, since repetition can shift a student into a different sixth-grade cohort than their actual treatment timing would imply. Second, I restrict the analysis to schools with consistently stable student populations across the three treated cohorts, requiring at least 80% of students to remain

in the same school from grade 1 to 5 in every cohort ($N = 4,765$ schools). This ensures that treatment assignment is tested only on schools where fifth-grade sector reliably reflects earlier grade 1 exposure. Because this individual-level tracking data is only available from 2017 onward, when the reform itself began, one concern is that school-level stability could itself be a consequence of treatment rather than a pre-existing characteristic (for instance, if smaller classes made families more likely to remain in the same school). I check this directly and find that school-level stability is equally flat across cohorts in non-priority schools, which were never exposed to the reform, as in REP+ and REP schools, suggesting that stability is a structural feature of these schools rather than a response to treatment. In both restricted samples, the results are consistent with those obtained in the main specification.

4 Empirical Strategy

Estimating the causal effect of class size reduction on student performance poses an identification challenge, as class size variation is typically endogenous, driven by family and school choices correlated with student outcomes. Recent literature addresses this endogeneity through experimental designs (Krueger, 1999; Banerjee et al., 2007) or quasi-experimental approaches exploiting discontinuities in class size rules (Angrist & Lavy, 1999; Hoxby, 2000; Fredriksson et al., 2013). I exploit a large-scale national reform in France to estimate the effects of class size reduction on student achievement. Beginning in 2017, the policy reduced first-grade class sizes to approximately 12 students in REP+ schools, extending to REP schools one year later, while non-priority schools remained untreated throughout. I use an event study design comparing the evolution of sixth-grade performance of REP+ and REP students with students from non-priority schools.

4.1 Event Study Specification

The analysis compares sixth-grade performance of students from REP+ or REP grade 5 schools (treated) with students from non-priority grade 5 schools (never treated). The first treated REP+ cohort entered grade 1 in 2017-2018 and grade 6 in September 2022; the first treated REP cohort entered grade 1 one year later and grade 6 in September 2023. Cohorts entering grade 6 before these respective dates were not yet exposed to the reform.

I estimate the following event study specification:

$$Y_{ist} = \alpha + \sum_{\substack{t=2017 \\ t \neq 2021}}^{2024} \tau_t (\text{Treated}_s \times \mathbb{1}[\text{Year}_t = t]) + \lambda_s + \delta_t + \epsilon_{ist} \quad (4.1)$$

where:

- Y_{ist} is the standardized test score (French or mathematics), or the binary readiness-to-succeed indicator, for student i from grade 5 school s entering grade 6 in year t
- Treated_s is an indicator for students from REP+ (or, in the corresponding specification, REP) grade 5 schools
- $t = 2021$ is the omitted reference year ($t = 2022$ for the REP comparison, given REP's later treatment date)
- λ_s are grade 5 school fixed effects
- δ_t are year fixed effects
- ϵ_{ist} is the error term

The coefficients τ_t capture the treatment effect in year t , the difference in outcomes between treated and non-priority students relative to the reference year 2021, conditional on school and year fixed effects. For $t < 2022$ (pre-treatment years), τ_t provides a test of the parallel trends assumption: these coefficients should be statistically indistinguishable from zero if treated and comparison groups followed similar trends before the reform. For $t \geq 2022$ (post-treatment years, or $t \geq 2023$ for the REP comparison), τ_t estimates the dynamic treatment effect, revealing whether impacts emerge immediately or evolve over time as additional treated cohorts enter grade 6.

School fixed effects λ_s absorb time-invariant differences between treated and non-priority schools, while year fixed effects δ_t control for common trends affecting all students, including the COVID-19 disruption in 2020. Standard errors are clustered at the grade 5 school level to account for within-school correlation in outcomes.

4.2 Identification

The event study estimator identifies causal effects under the parallel trends assumption: absent the reform, treated and non-priority students would have experienced the same evolution in grade 6 achievement. Formally, this requires:

$$\begin{aligned} & \mathbb{E}[Y_{ist}(0)|\text{Treated}_s = 1, t] - \mathbb{E}[Y_{ist}(0)|\text{Treated}_s = 1, t - 1] \\ &= \mathbb{E}[Y_{ist}(0)|\text{Treated}_s = 0, t] - \mathbb{E}[Y_{ist}(0)|\text{Treated}_s = 0, t - 1] \end{aligned} \quad (4.2)$$

where $Y_{ist}(0)$ denotes potential outcomes in the absence of treatment.

I assess this assumption graphically by plotting the event study coefficients τ_t over time, examining whether pre-treatment coefficients are close to zero and statistically insignificant, and formally through joint F-tests of the pre-treatment coefficients, reported alongside the main results in Section 5.1 and summarized in Table A.1.

5 Analysis

This section examines three outcomes: mathematics and French test scores, and students' self-reported readiness to succeed. The primary comparison is between students from REP+ schools, whose first treated cohort entered grade 6 in 2022, and students from non-priority schools, which never received treatment. This comparison offers the longest post-treatment window, covering three treated cohorts entering sixth grade in 2022, 2023, and 2024. I also report estimates comparing students from REP schools, treated one year later than REP+, to students from non-priority schools. A third comparison, between REP+ and REP directly, is discussed in the appendix (Section A.5); because REP itself becomes treated in 2023, this comparison does not allow for a dynamic analysis. Robustness checks addressing student mobility and grade retention are reported in Appendix Section A.4.

5.1 Effects on Mathematics

Figure 5.1 shows the evolution, from 2017 to 2024, of average standardized mathematics scores at grade 6 entry, for students originating from REP+, REP, and non-priority grade 5

schools. Although REP+ and REP score persistently below non-priority schools, all three groups follow a similar evolution through 2019, before a sharp decline for REP+ and REP in 2020, while scores in non-priority schools remain comparatively stable. This divergence coincides with the school closures of March 2020, which affected the cohort entering sixth grade that September during their fifth grade, a critical year for consolidating the skills tested at sixth-grade entry. DEPP studies documented that this disruption disproportionately affected disadvantaged students, more heavily represented in REP and REP+ schools (DEPP, 2022), consistent with the sharper decline observed here for these two groups.

Figure 5.1 also plots a dashed line connecting the 2019 and 2021 observations for REP and REP+, bypassing 2020. Both lines return closely to their pre-pandemic trajectory by 2021, indicating that the 2020 disruption was a temporary shock rather than a lasting shift in the relative performance of REP and REP+ schools.

Because each event study coefficient τ_t is estimated relative to 2021, the estimates for 2022, 2023, and 2024 do not incorporate 2020 data directly; the 2020 coefficient is reported separately as a diagnostic of the pandemic disruption. The pooled two-way fixed effects estimates reported below also control for the 2020 disruption directly.

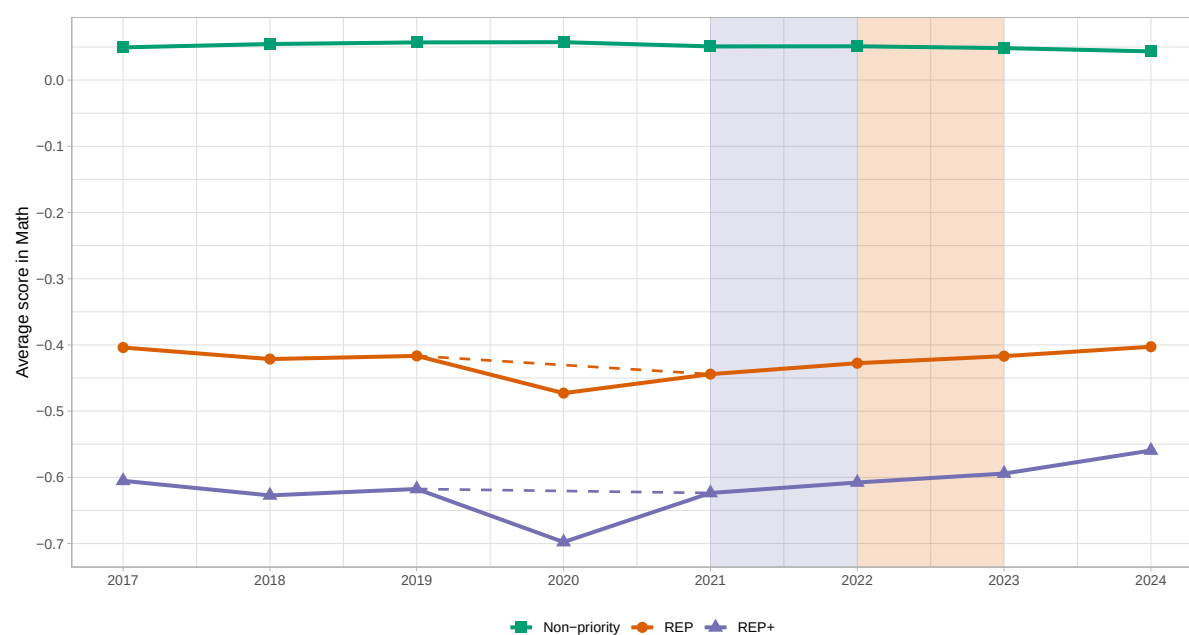


Figure 5.1: Evolution of the standardized score in math, 2017-2024

Source: DEPP, National assessments at sixth-grade entry, 2017-2024. Sample includes all public and private schools in France. Shaded regions indicate reform implementation: purple (2021-2022) marks the transition to the first REP+ cohort treated in grades 1-2; orange (2022-2023) marks the same transition for REP schools. Dashed lines connect 2019 to 2021 for REP and REP+, bypassing the 2020 disruption.

REP+ vs Non-priority

Figure 5.2 presents event study estimates comparing REP+ and non-priority schools. Although the joint test of parallel trends over 2017-2019 is rejected ($F = 4.49$, $p = 0.004$, Table A.1), individual pre-treatment coefficients are small in magnitude and none is individually significant. Restricting the sample to schools with consistently stable student populations across cohorts (Section A.4) removes this violation entirely ($F = 0.90$, $p = 0.44$), while leaving the pattern of results essentially unchanged, indicating that the joint test in the main specification reflects student mobility rather than a substantive threat to identification.

Post-treatment estimates are positive throughout and grow markedly across cohorts: 0.019 standard deviations for the first treated cohort in 2022 ($p < 0.05$), 0.036 standard deviations for the second cohort in 2023 ($p < 0.01$), and 0.070 standard deviations for the third cohort in 2024 ($p < 0.01$). The pooled two-way fixed effects estimate, averaging across the three post-treatment cohorts, is 0.043 standard deviations ($p < 0.01$, Appendix Table A.4).

This growing pattern is confirmed by a formal sensitivity analysis. Using the relative-magnitudes approach of Rambachan and Roth (2023), the 2024 estimate remains statistically distinguishable from zero even when allowing the post-treatment counterfactual trend to deviate by up to 0.55 times the largest observed pre-treatment violation (Figure 5.3), a substantially more generous allowance than the pre-treatment coefficients themselves would suggest is warranted. The estimate is therefore not attributable to a continuation of pre-existing differential trends.

Taken together, these results indicate a positive effect of the reform on mathematics achievement that is small in the first treated cohort, but grows into a substantial and well-identified effect by the third cohort, consistent with a reform whose impact strengthens as schools become more familiar with reduced class sizes and adapt their teaching practices accordingly.

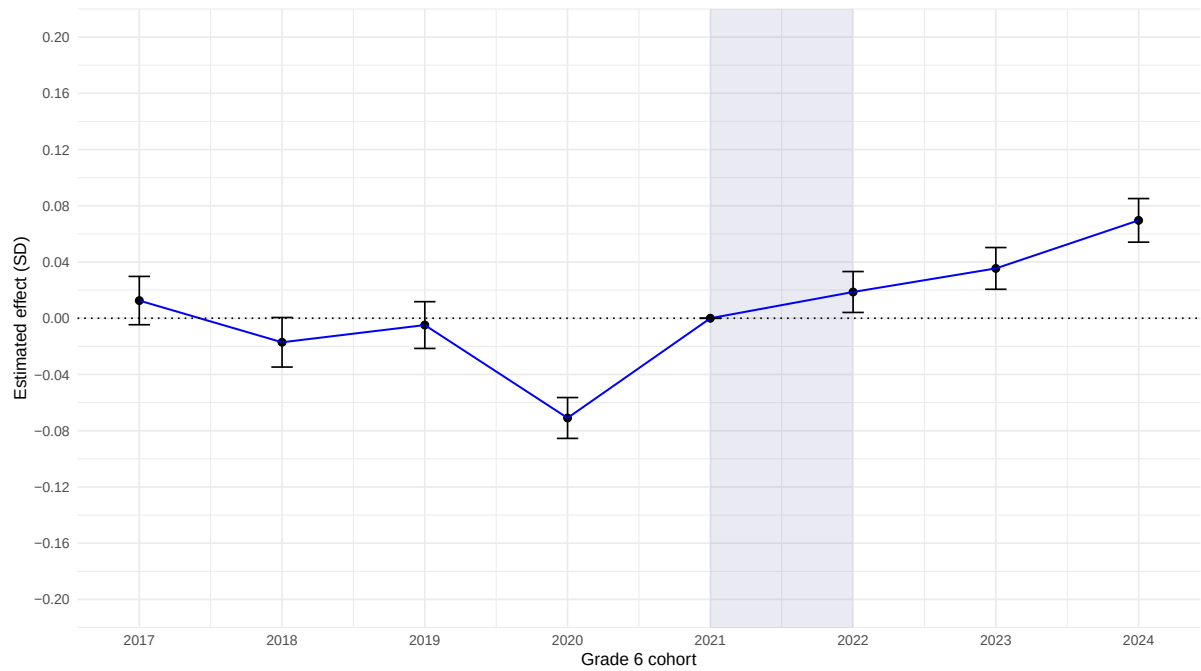


Figure 5.2: Event study estimates: Math score, REP+ vs non-priority

Note: Event study coefficients τ_t from equation (1), no student-level controls. Standard errors clustered at the grade 5 school level. 95% confidence intervals displayed.

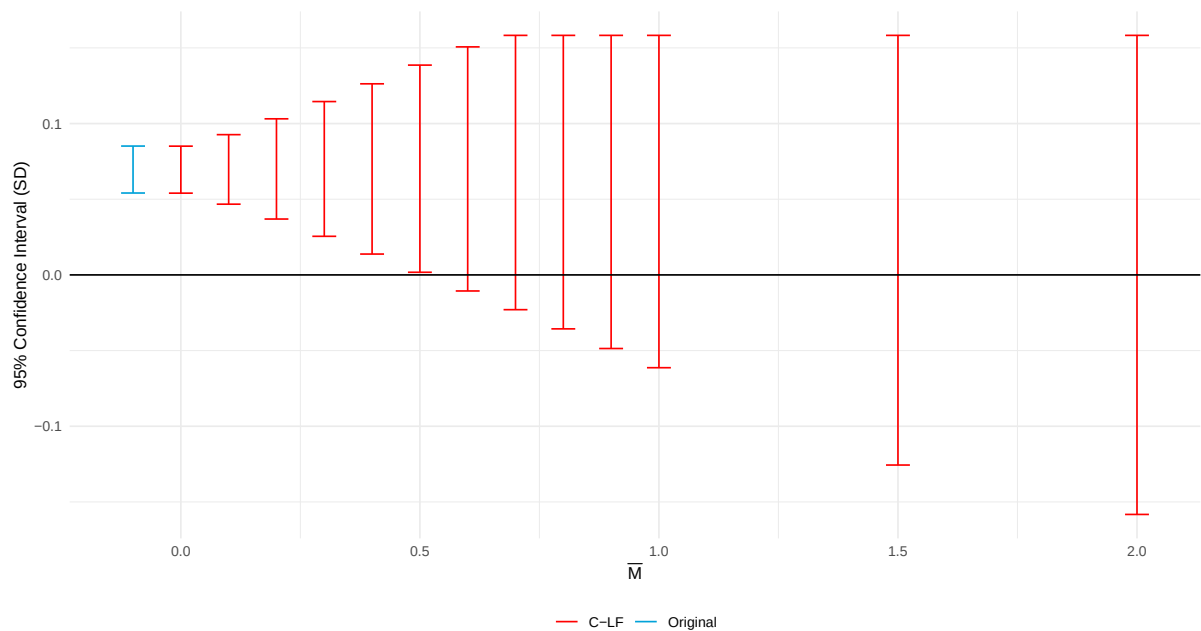


Figure 5.3: Sensitivity analysis: Math score, REP+ vs non-priority, 2024

Note: Robust confidence intervals under the relative-magnitudes restriction of Rambachan and Roth (2023), for values of $\bar{M}$ from 0 to 2. The blue interval is the original, unrestricted confidence interval.

REP vs Non-priority

Figure 5.4 presents the corresponding estimates comparing REP and non-priority schools. Individual pre-treatment coefficients remain small in magnitude, and post-treatment estimates follow the same pattern found for REP+: 0.015 standard deviations for the first treated cohort in 2023 ($p < 0.05$), and 0.032 standard deviations for the second cohort in 2024 ($p < 0.01$). These results corroborate the findings for REP+, pointing to the same pattern of a growing effect on mathematics achievement across cohorts.

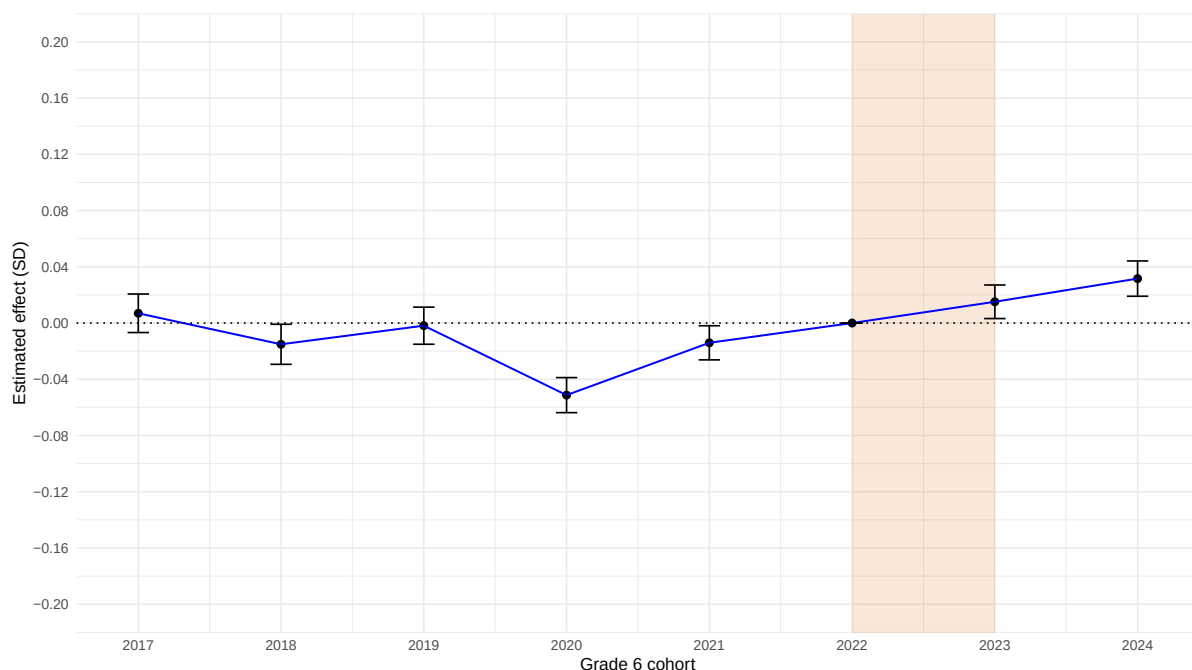


Figure 5.4: Event study estimates: Math score, REP vs non-priority

Note: Event study coefficients τ_t from equation (1), no student-level controls. Standard errors clustered at the grade 5 school level. 95% confidence intervals displayed.

5.2 Effects on French

Figure 5.5 shows the evolution, from 2017 to 2024, of average standardized French scores at grade 6 entry, for students originating from REP+, REP, and non-priority grade 5 schools. Unlike mathematics, REP+ already shows some divergence from non-priority schools as early as 2017, consistent with the pre-trend violation documented below. As with mathematics, a sharp decline is visible for REP+ and REP in 2020 alongside the school closures documented above, while scores in non-priority schools remain comparatively stable.

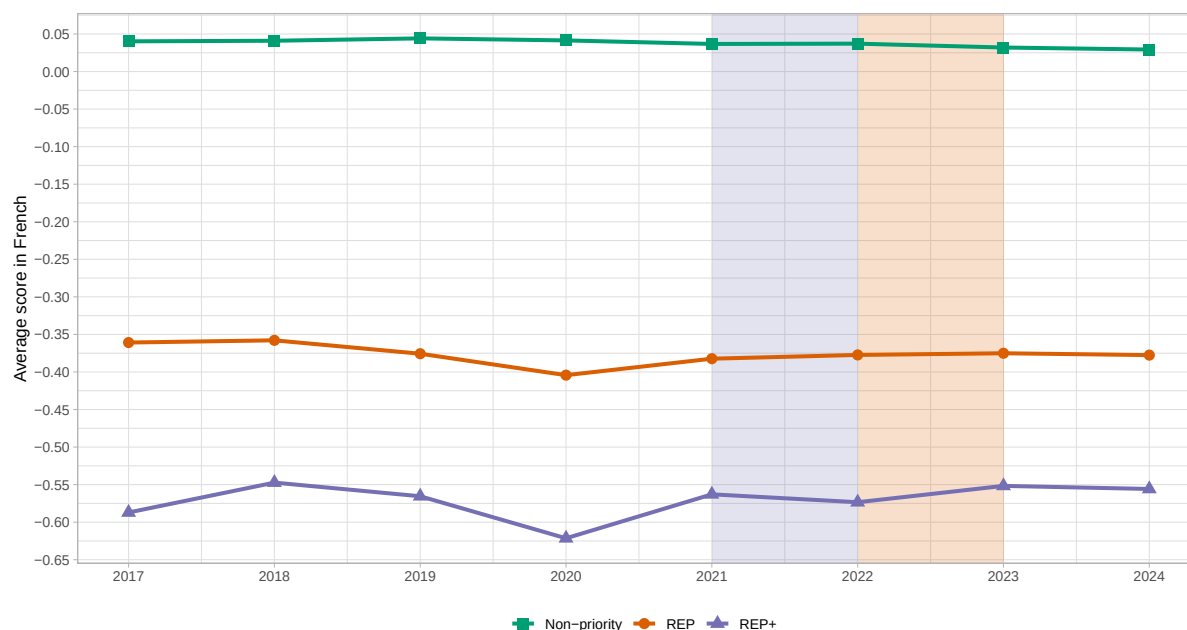


Figure 5.5: Evolution of the standardized score in French, 2017-2024

Source: DEPP, National assessments at sixth-grade entry, 2017-2024. Sample includes all public and private schools in France. Shaded regions indicate reform implementation: purple (2021-2022) marks the transition to the first REP+ cohort treated in grades 1-2; orange (2022-2023) marks the same transition for REP schools.

REP+ vs Non-priority

Figure 5.6 presents event study estimates comparing REP+ and non-priority schools. The joint test of parallel trends over 2017-2019 is rejected ($F = 7.28$, $p < 0.001$, Table A.1), driven by negative and significant coefficients in 2017 and, to a lesser extent, 2019. Post-treatment estimates are small throughout: no effect is detectable for the first treated cohort in 2022, a small statistically significant effect of 0.017 standard deviations emerges for the second cohort in 2023 ($p < 0.05$), and the estimate for the third cohort in 2024 is not statistically significant. The pooled two-way fixed effects estimate is 0.021 standard deviations (Appendix Table A.4), not statistically distinguishable from zero.

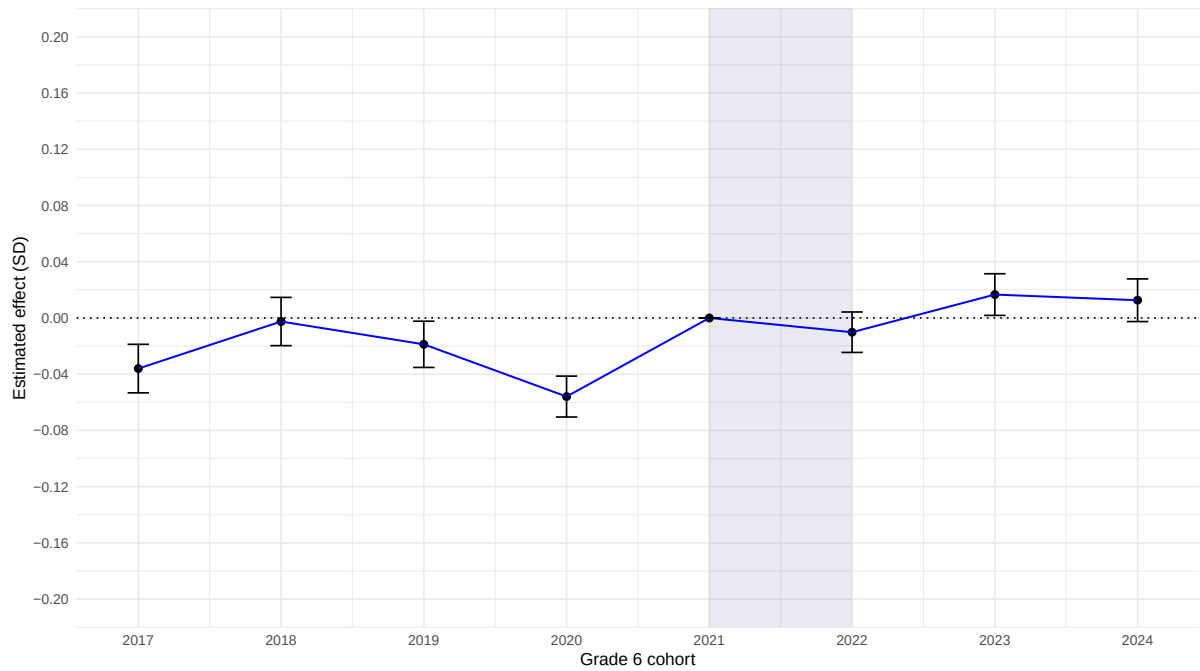


Figure 5.6: Event study estimates: French score, REP+ vs non-priority

Note: Event study coefficients τ_t from equation (1). Standard errors clustered at the grade 5 school level. 95% confidence intervals displayed.

REP vs Non-priority

Figure 5.7 presents the corresponding estimates comparing REP and non-priority schools. Here, the joint test of parallel trends cannot be rejected ($F = 1.40$, $p = 0.231$), the only specification among those considered in which pre-treatment coefficients are jointly indistinguishable from zero. Post-treatment estimates remain small and not statistically significant in either year (0.008 standard deviations in 2023; 0.006 standard deviations in 2024). Where parallel trends hold most convincingly, the estimated effect remains null, reinforcing the conclusion that, unlike mathematics, the class size reduction had no lasting effect on French achievement.

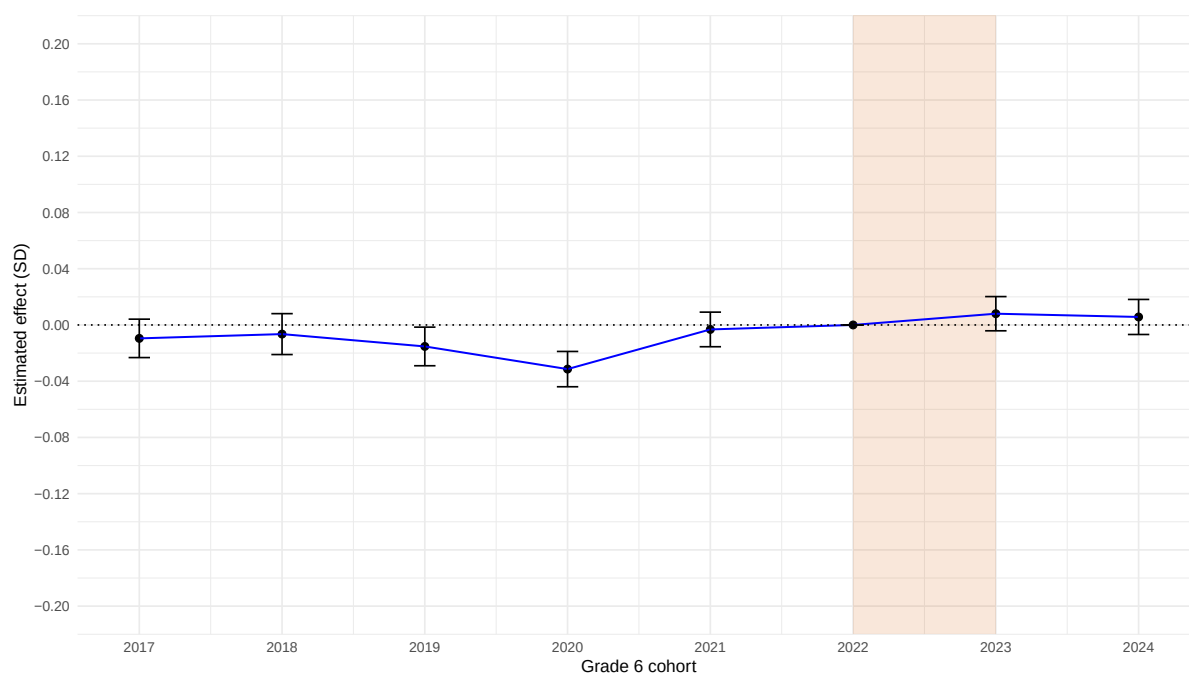


Figure 5.7: Event study estimates: French score, REP vs non-priority

Note: Event study coefficients τ_t from equation (1). Standard errors clustered at the grade 5 school level. 95% confidence intervals displayed.

5.3 Effects on Readiness to Succeed

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Since 2020, sixth-grade assessments have included a question measuring students' self-confidence: "At the beginning of this sixth-grade year, do you think you are ready to succeed in your year?" This question captures a prospective belief in one's own ability to succeed, consistent with the concept of self-efficacy described by Duckworth and Yeager (2015). I construct a binary outcome equal to one if the student responds positively ("Yes" or "Yes, absolutely") and zero otherwise.

Figure 5.8 shows the evolution, from 2020 to 2024, of the share of students reporting readiness to succeed at grade 6 entry, for students originating from REP+, REP, and non-priority grade 5 schools. Readiness rises across all three groups over the period, with REP schools showing a particularly marked increase from 2020 onward.

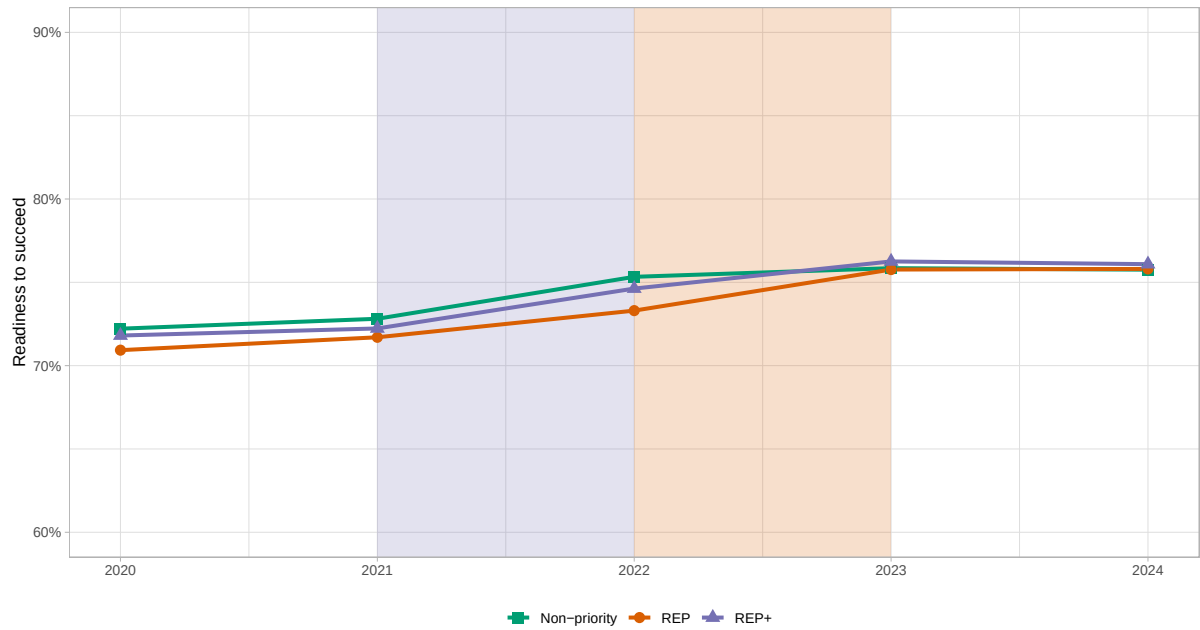


Figure 5.8: Evolution of Self-Reported Readiness to Succeed, 2020-2024

Source: DEPP, National assessments at sixth-grade entry, 2020-2024. Sample includes all public and private schools in France. Shaded regions indicate reform implementation: purple (2021-2022) marks the transition to the first REP+ cohort treated in grades 1-2; orange (2022-2023) marks the same transition for REP schools.

REP+ vs Non-priority

Because this question was introduced in 2020, only one pre-treatment year is available for this comparison. The corresponding coefficient is small and not statistically significant (0.002, $p = 0.60$), and readiness to succeed does not show the differential COVID-19 disruption observed for test scores.

Figure 5.9 presents event study estimates. No effect is detectable for the first treated cohort in 2022, but small, statistically significant effects emerge for the second and third cohorts: a 1.05 percentage point increase in 2023 ($p < 0.05$) and a 0.99 percentage point increase in 2024 ($p < 0.05$). The pooled two-way fixed effects estimate is 0.6 percentage points ($p < 0.05$, Appendix Table A.6). This delayed emergence mirrors the pattern observed for mathematics.

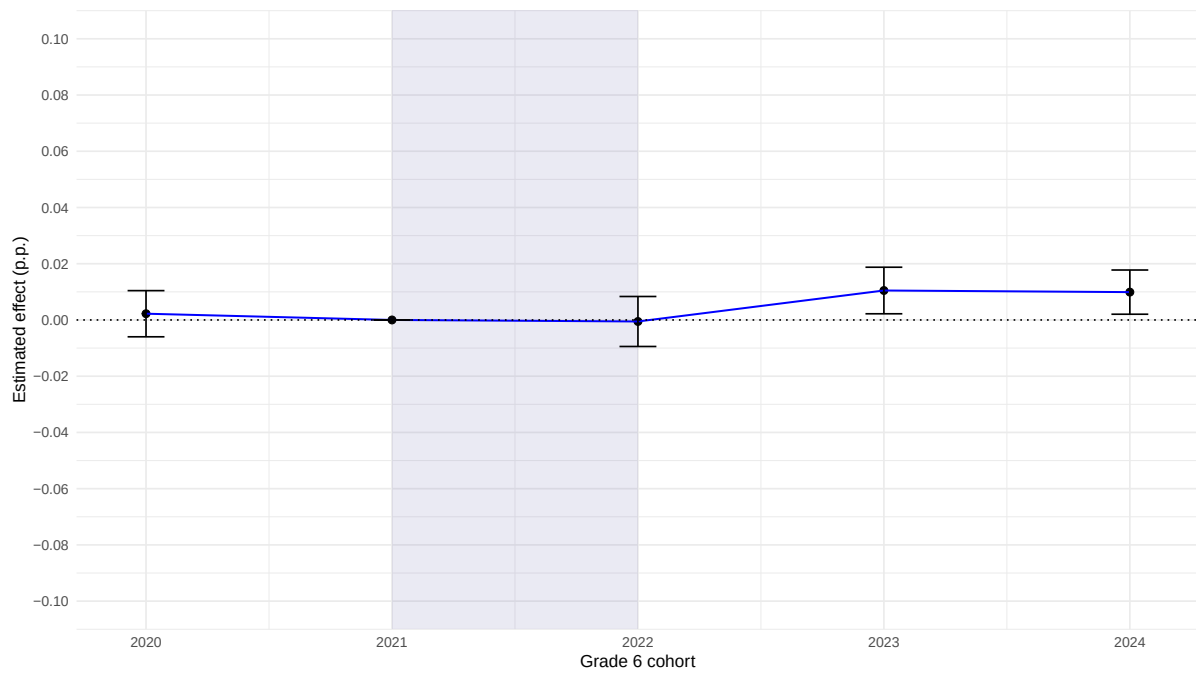


Figure 5.9: Event study estimates: Readiness to succeed, REP+ vs non-priority

Note: Event study coefficients τ_t from equation (1) using the binary readiness-to-succeed indicator. Coefficients represent percentage point changes. Standard errors clustered at the grade 5 school level. 95% confidence intervals displayed.

REP vs Non-priority

Figure 5.10 presents the corresponding estimates comparing REP and non-priority schools. Both available pre-treatment coefficients are positive and statistically significant (2020: 0.8 percentage points, $p < 0.05$; 2021: 0.9 percentage points, $p < 0.05$), and the joint test of parallel trends is rejected ($F = 3.59$, $p = 0.028$). Post-treatment estimates are larger than for REP+: a 2.00 percentage point increase in 2023 ($p < 0.01$) and a 2.08 percentage point increase in 2024 ($p < 0.01$). Given the pre-treatment violation, I treat the REP+ estimate as the more credible measure of the reform's effect on readiness to succeed.

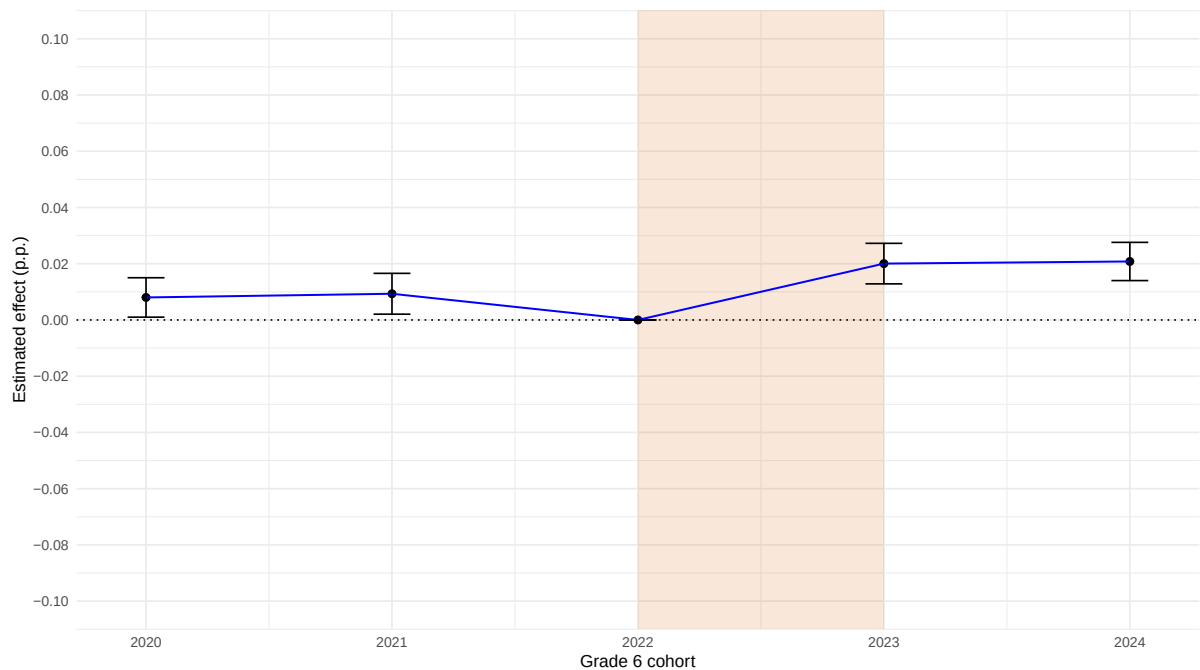


Figure 5.10: Event study estimates: Readiness to succeed, REP vs non-priority

Note: Event study coefficients τ_t from equation (1) using the binary readiness-to-succeed indicator. Coefficients represent percentage point changes. Standard errors clustered at the grade 5 school level. 95% confidence intervals displayed.

6 Discussion and Conclusion

6.1 Summary of Findings

This study estimates the long-term effects of France’s large-scale class size reduction reform on sixth-grade achievement and non-cognitive outcomes. Using administrative data covering all students entering sixth grade between 2017 and 2024, I employ an event study design comparing REP+ students, first treated in grade 1 in 2017-2018, with students from non-priority schools, never treated. I corroborate these estimates using a second comparison between REP students, treated one year later, and non-priority schools.

The results point to a positive effect on mathematics achievement, and no detectable effect on French. For mathematics, the estimated effect for the first treated cohort in 2022 is small and does not survive robustness checks, but a clear effect emerges by the second treated cohort and grows substantially by the third, reaching 0.07 standard deviations in 2024. This growing pattern is robust to excluding grade repeaters from the sample and to restricting the sample to schools where student populations remained stable across cohorts,

addressing concerns that student mobility could otherwise distort treatment assignment. It also survives a formal sensitivity analysis allowing for substantial deviations from parallel pre-trends, confirming that it reflects a strengthening effect of the reform rather than pre-existing trends. For French, no statistically significant effect is detected using either comparison group, and where parallel pre-trends hold most convincingly (the REP vs non-priority comparison) the estimated effect remains null. The reform is also associated with a small but statistically significant increase in students' self-reported readiness to succeed at sixth-grade entry, following the same delayed pattern observed for mathematics.

These long-term effects remain smaller overall than the short-term impacts documented by Andreu et al. (2021), who found effects of 0.14 standard deviations in French and 0.12 in mathematics at the end of first grade, using a representative sample comparing REP+ to REP schools. Notably, this short-term evaluation concerned the first treated cohort that started grade 1 in 2017. A closer examination of the long-term effect on this cohort when they enter grade 6 in 2022, using the same REP+ vs REP comparison (El Bah, 2026), finds null effects in both subjects at sixth-grade entry, suggesting that the short-term gains observed at the end of first and second grade had substantially faded by the time this cohort reached middle school. The results reported in this paper are consistent with this conclusion: the REP+ vs non-priority estimate for mathematics in 2022, though positive and statistically significant in the main specification, does not survive the same robustness checks that support the effects documented for later cohorts. Taken together, these findings point to the same conclusion for the first treated cohort: the short-term gains documented by Andreu et al. (2021) had faded to a null effect on both subjects by sixth-grade entry.

The subsequent, second and third treated cohorts, by contrast, show growing and statistically significant effects on mathematics that do survive these robustness checks. As discussed in Section 3.3, treatment assignment based on students' fifth-grade school is subject to measurement error from grade retention and mobility between the first and fifth grade, both of which are expected to bias estimates toward zero; the estimates reported here should accordingly be read as intent-to-treat effects that likely understate the effect of actually experiencing smaller classes throughout grades 1-2. This is particularly relevant for interpreting the gap between the short-term and long-term estimates: part of the

apparent gap between the short-term gains of 0.12-0.14 standard deviations and the long-term mathematics effect of 0.07 standard deviations by the third cohort may reflect this attenuation, and the true effect for later cohorts may be closer in magnitude to the short-term estimates than the reported figures suggest. Despite this attenuation, the mathematics effect stands out as a robust result: unlike French, it grows across successive treated cohorts, suggesting that the benefits of smaller classes for mathematics achievement strengthen as the reform becomes more established within schools.

6.2 Interpretation and Mechanisms

The pattern documented here is consistent with an implementation story rather than a purely mechanical fade-out of short-term gains. The first year of the reform, 2017-2018, was a period of adjustment: IGÉSR (2023) documents space constraints and varied implementation modalities during the initial rollout, including co-teaching arrangements and classes sharing rooms, which may have limited the reform's effectiveness for the first treated cohort specifically. The short-term evaluation also found that teachers did not systematically change their pedagogical practices in reduced-size classes during this initial period (Andreu et al., 2021), consistent with evidence from other contexts showing that teacher adaptation to smaller classes can take time (Finn & Shanahan, 2016). As implementation stabilized, it is possible that instructional practices adapted, allowing the effects observed for later cohorts to be sustained through sixth grade rather than fading as they appear to have done for the first cohort.

Because French and mathematics are typically taught by the same teacher in primary schools, this adaptation story does not on its own explain why the sustained effect emerges specifically in mathematics rather than French. One possible explanation for this subject-specific pattern lies not in the students' skill acquisition but in how much of each subject's achievement is actually produced by schools. A well-documented finding in the education literature is that mathematics achievement is disproportionately produced during the school year, since most families provide comparatively little math input or practice at home, whereas reading and language skills continue to develop outside school, supported by everyday family reading, conversation, and language exposure (Downey, von Hippel, & Broh, 2004). Under this view, school-based interventions such as class size reduction have

more room to influence mathematics achievement specifically, since family inputs already substantially shape French achievement and may partly offset or dilute the effect of a school-level intervention on this subject. This may help explain why a sustained effect on mathematics, but not French, emerges as the reform becomes established.

6.3 Avenues for Future Research

A first avenue concerns the extension of class size reduction to kindergarten in REP and REP+ schools, which began in 2020. Because this extension allows some students to experience three consecutive years of reduced-size classes rather than two, it offers a natural opportunity to examine whether earlier and more intensive exposure produces larger and more persistent effects on achievement, consistent with Heckman's (2006) work on the returns to early childhood investment.

A second avenue is to continue following these cohorts as they progress through middle school and beyond, both to assess whether the mathematics effect documented here at sixth-grade entry is sustained over time, and to examine whether class size reduction affects other dimensions of student outcomes beyond test scores, such as tracking decisions after ninth grade, trajectories into secondary and higher education, and non-cognitive outcomes beyond the readiness-to-succeed measure examined here. Understanding whether these effects persist, strengthen, or fade further as students advance, and whether they extend to outcomes beyond academic achievement, would meaningfully sharpen the long-term picture presented here.

Finally, this study has focused entirely on student outcomes, but the reform's effects may extend beyond the students it directly targets. Teacher mobility is particularly high in disadvantaged schools (Prost, 2012; Loeb et al., 2012), and the short-term evaluation found that REP+ teachers in reduced first-grade classes reported greater confidence, more serene classroom climates, and more favorable working conditions (Blouet et al., 2020). Investigating whether these improved working conditions lead to higher teacher retention and stability in priority education schools would help determine whether the policy generated an additional, unintended benefit for the teaching workforce, alongside its direct effects on students.

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Appendices

A Additional Results and Robustness Checks

A.1 Pre-Trend F-Tests

Table A.1: Pre-Trend F-Tests

Specification	Comparison	Outcome	F-statistic	p-value
Main	REP+ vs Non-priority	Math	4.49	0.004
		French	7.28	<0.001
		Readiness	0.28	0.595
	REP vs Non-priority	Math	4.03	0.003
		French	1.40	0.231
		Readiness	3.59	0.028
Robustness: on-time students	REP+ vs Non-priority	Math	5.99	<0.001
		French	3.00	0.029
Robustness: stable schools	REP+ vs Non-priority	Math	0.90	0.438
		French	5.48	<0.001

Note: F-tests are joint tests of the significance of pre-treatment event-study coefficients (2017-2019 for the REP+ vs Non-priority comparison; 2017-2019 and 2021 for the REP vs Non-priority comparison, given REP's later treatment date). The COVID-19 year (2020) is excluded from all pre-trend tests. For readiness to succeed, only one pre-treatment coefficient is available for the REP+ vs Non-priority comparison (2020), so the reported F-test is equivalent to a two-tailed t-test on that single coefficient.

A.2 Event Study Estimations

Table A.2: Event Study Estimates: French and Math Scores

	REP+ vs Non-priority		REP vs Non-priority	
	French	Mathematics	French	Mathematics
	(1)	(2)	(3)	(4)
Treatment $\times$ 2017	−0.0360*** (0.0088)	0.0126 (0.0088)	−0.0095 (0.0070)	0.0069 (0.0070)
Treatment $\times$ 2018	−0.0025 (0.0088)	−0.0171* (0.0090)	−0.0065 (0.0074)	−0.0151** (0.0073)
Treatment $\times$ 2019	−0.0188** (0.0084)	−0.0048 (0.0085)	−0.0153** (0.0070)	−0.0019 (0.0067)
Treatment $\times$ 2020	−0.0559*** (0.0074)	−0.0709*** (0.0074)	−0.0314*** (0.0064)	−0.0513*** (0.0064)
Treatment $\times$ 2021			−0.0032 (0.0063)	−0.0140** (0.0062)
Treatment $\times$ 2022	−0.0101 (0.0073)	0.0187** (0.0074)		
Treatment $\times$ 2023	0.0166** (0.0076)	0.0355*** (0.0076)	0.0080 (0.0062)	0.0151** (0.0061)
Treatment $\times$ 2024	0.0126 (0.0077)	0.0696*** (0.0079)	0.0057 (0.0064)	0.0316*** (0.0064)
School FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	4,538,986	4,541,026	4,772,675	4,774,573
R ²	0.10452	0.11930	0.09626	0.11147

Note: Dependent variable is standardized test score (mean 0, SD 1). Reference year is 2021 for REP+ vs Non-priority, 2022 for REP vs Non-priority. Standard errors clustered at the grade 5 school level in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Table A.3: Event Study Estimates: Readiness to Succeed

	REP+ vs Non-priority (1)	REP vs Non-priority (2)
Treatment $\times$ 2020	0.0022 (0.0042)	0.0080** (0.0036)
Treatment $\times$ 2021		0.0093** (0.0037)
Treatment $\times$ 2022	-0.0005 (0.0045)	
Treatment $\times$ 2023	0.0105** (0.0042)	0.0200*** (0.0037)
Treatment $\times$ 2024	0.0099** (0.0040)	0.0208*** (0.0035)
School FE	Yes	Yes
Year FE	Yes	Yes
Observations	1,846,416	1,935,670
R ²	0.02091	0.02091

Note: Dependent variable is a binary readiness-to-succeed indicator. Coefficients represent percentage point changes. Reference year is 2021 for REP+ vs Non-priority, 2022 for REP vs Non-priority. Standard errors clustered at the grade 5 school level in parentheses. ***p<0.01, **p<0.05, *p<0.1.

A.3 Pooled Estimates

Table A.4: Pooled Estimates: REP+ vs Non-priority

	French (1)	Mathematics (2)
Treatment $\times$ Post	0.0207*** (0.0049)	0.0432*** (0.0052)
Treatment $\times$ 2020	-0.0416*** (0.0062)	-0.0687*** (0.0062)
School FE	Yes	Yes
Year FE	Yes	Yes
Observations	4,538,986	4,541,026
R ²	0.10451	0.11928

Note: Dependent variable is standardized test score. This specification pools all post-treatment years (2022-2024). Treatment $\times$ 2020 accounts for the differential impact of school closures on REP+ students in 2020. Standard errors clustered at the grade 5 school level in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Table A.5: Pooled Estimates: REP vs Non-priority

	French (1)	Mathematics (2)
Treatment $\times$ Post	0.0137*** (0.0042)	0.0281*** (0.0044)
Treatment $\times$ 2020	-0.0245*** (0.0050)	-0.0464*** (0.0050)
School FE	Yes	Yes
Year FE	Yes	Yes
Observations	4,772,675	4,774,573
R ²	0.09626	0.11146

Note: Dependent variable is standardized test score. This specification pools all post-treatment years (2023-2024). Treatment $\times$ 2020 accounts for the differential impact of school closures on REP students in 2020. Standard errors clustered at the grade 5 school level in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Note on French: The pooled estimate for French is significant but the corresponding event study estimates (Table A.2) show no statistically significant effect in either individual post-treatment year (2023 or 2024).

Table A.6: Pooled Estimates: Readiness to Succeed

	REP+ vs Non-priority (1)	REP vs Non-priority (2)
Treatment $\times$ Post	0.0063** (0.0026)	0.0140*** (0.0021)
School FE	Yes	Yes
Year FE	Yes	Yes
Observations	1,846,416	1,935,670
R ²	0.02090	0.02090

Note: Dependent variable is a binary readiness-to-succeed indicator. This specification pools all post-treatment years. Readiness to succeed does not show a differential COVID-19 disruption for either comparison, so no COVID-19 control is included. Standard errors clustered at the grade 5 school level in parentheses. ***p<0.01, **p<0.05, *p<0.1.

A.4 Sample Restrictions

This section reports two robustness checks addressing the sources of measurement error in treatment assignment discussed above.

The first restricts the sample to on-time students. A student is considered *late* if they are at least one year older than expected for their grade level, based on their date of birth and grade level at sixth-grade entry; this age-based indicator serves as a proxy for grade retention at any point in a student's schooling history. Excluding late students addresses the possibility that grade retention, which can shift a student into a different sixth-grade cohort than their actual treatment timing would imply, distorts the estimated pattern of results across cohorts.

The second restricts the sample to schools with consistently stable student populations. Using DIAPRE, I track three cohorts of students entering grade 1 in 2017, 2018, and 2019 through grade 5, and compute, for each school, the share of students remaining in the same school from grade 1 to grade 5 in each cohort. Schools are classified as stable if at least 80% of students remain in the same school from grade 1 to grade 5 in every one of the three cohorts ($N = 4,765$ schools). Because treatment assignment is based on students' grade 5 school, this restriction tests whether the main results hold in the subset of schools where grade 5 sector most reliably reflects earlier grade 1-2 exposure to the reform.

Results from both restricted samples are reported below.

A.4.1 On-Time Students

Table A.7: Robustness: Event Study Estimates, On-Time Students (REP+ vs Non-priority)

	French (1)	Mathematics (2)
Treatment $\times$ 2017	−0.0217** (0.0091)	0.0276*** (0.0092)
Treatment $\times$ 2018	0.0037 (0.0092)	−0.0059 (0.0094)
Treatment $\times$ 2019	−0.0094 (0.0087)	−0.0017 (0.0089)
Treatment $\times$ 2020	−0.0540*** (0.0077)	−0.0695*** (0.0077)
Treatment $\times$ 2022	−0.0084 (0.0077)	0.0158** (0.0077)
Treatment $\times$ 2023	0.0095 (0.0079)	0.0275*** (0.0079)
Treatment $\times$ 2024	0.0059 (0.0081)	0.0636*** (0.0083)
School FE	Yes	Yes
Year FE	Yes	Yes
Observations	4,254,656	4,257,105
R ²	0.10119	0.11685

Note: Sample restricted to on-time students. Reference year is 2021. Standard errors clustered at the grade 5 school level in parentheses. ***p<0.01, **p<0.05, *p<0.1.

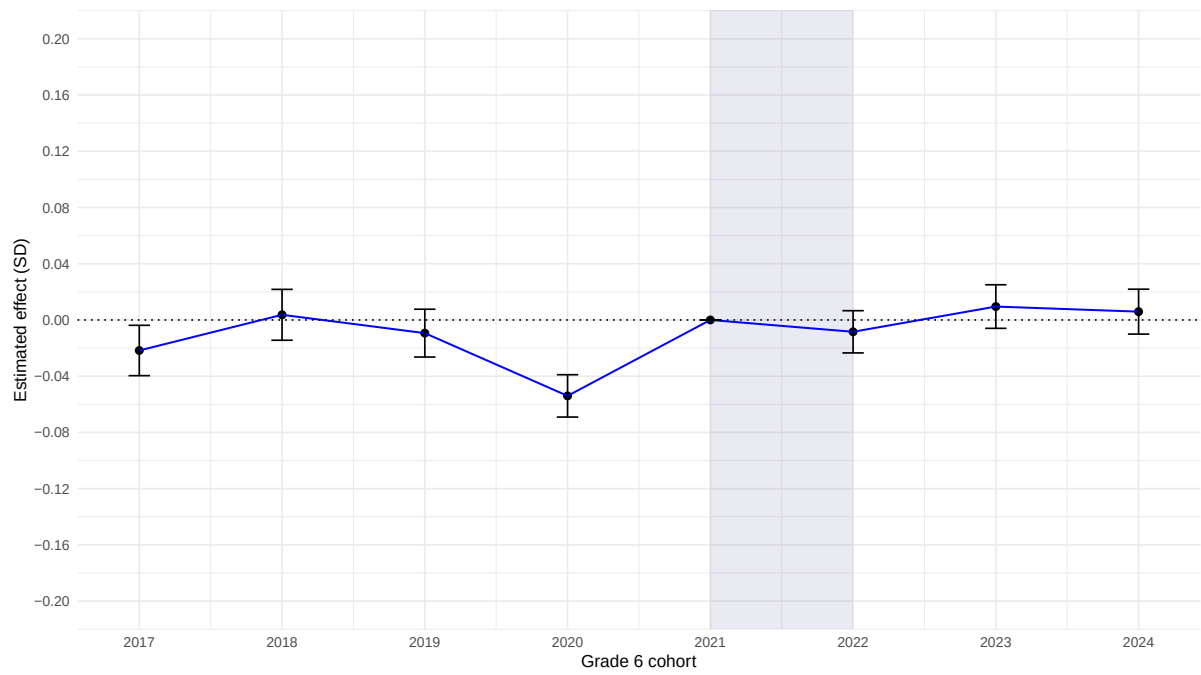


Figure A.1: Event study estimates: French score, REP+ vs Non-priority (on-time students)

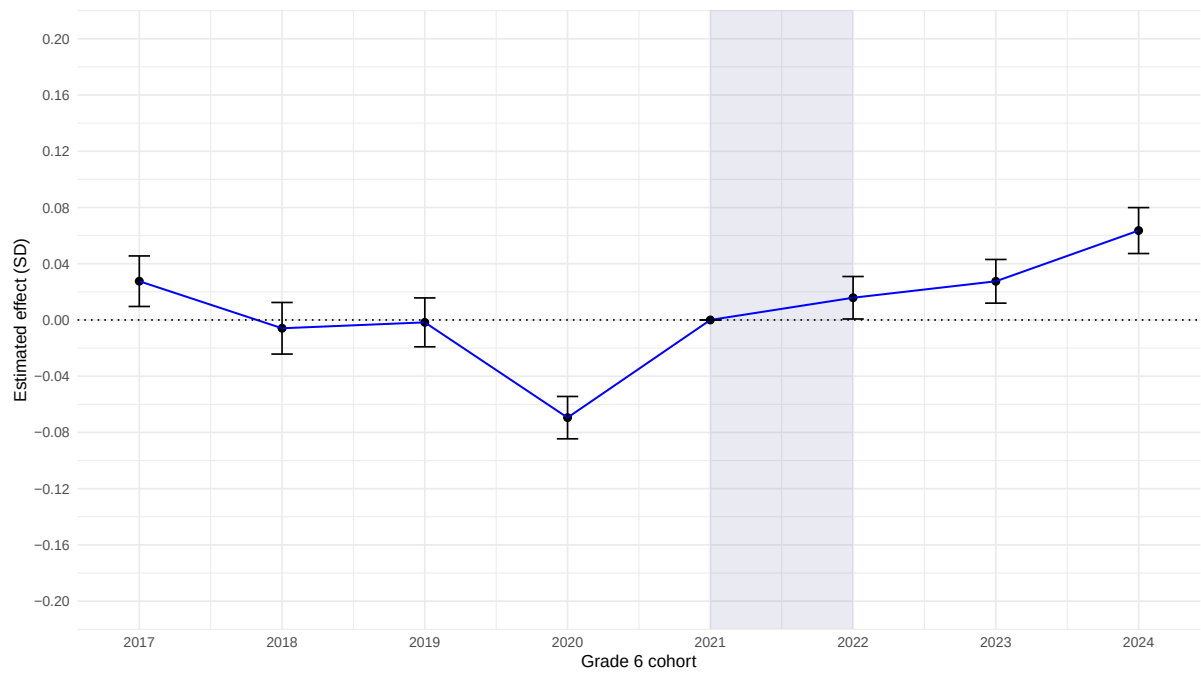


Figure A.2: Event study estimates: Math score, REP+ vs Non-priority (on-time students)

A.4.2 Stable Schools

Table A.8: Robustness: Event Study Estimates, Stable Schools (REP+ vs Non-priority)

	French	Mathematics
	(1)	(2)
Treatment $\times$ 2017	−0.0519** (0.0202)	−0.0132 (0.0211)
Treatment $\times$ 2018	0.0313 (0.0204)	−0.0003 (0.0215)
Treatment $\times$ 2019	−0.0190 (0.0211)	−0.0295 (0.0215)
Treatment $\times$ 2020	−0.0597*** (0.0169)	−0.0923*** (0.0173)
Treatment $\times$ 2022	−0.0056 (0.0168)	0.0145 (0.0185)
Treatment $\times$ 2023	0.0216 (0.0183)	0.0348* (0.0182)
Treatment $\times$ 2024	−0.0051 (0.0183)	0.0601*** (0.0186)
School FE	Yes	Yes
Year FE	Yes	Yes
Observations	647,850	648,452
R ²	0.13187	0.14583

Note: Sample restricted to schools with at least 80% of students remaining in the same school from grade 1 to grade 5 in each of three tracked cohorts (N = 4,765 schools). Reference year is 2021. Standard errors clustered at the grade 5 school level in parentheses. ***p<0.01, **p<0.05, *p<0.1.

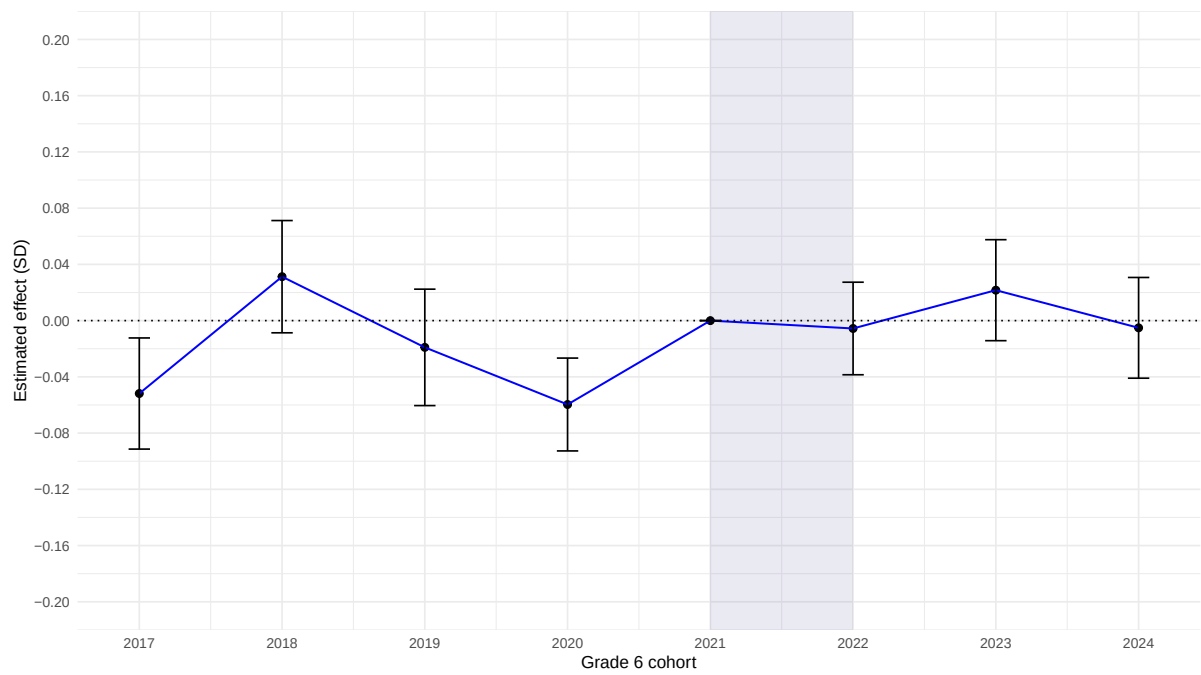


Figure A.3: Event study estimates: French score, REP+ vs Non-priority (stable schools)

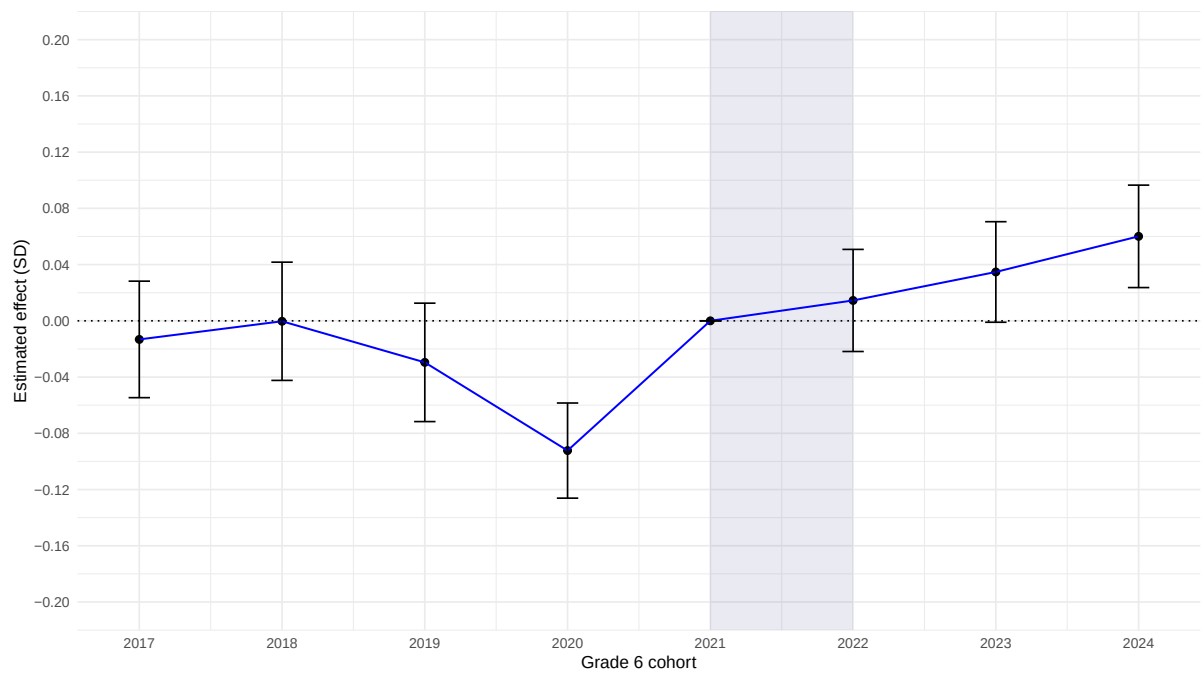


Figure A.4: Event study estimates: Math score, REP+ vs Non-priority (stable schools)

A.5 REP+ vs REP

A.5.1 Pre-trend F-test

Table A.9: REP+ vs REP: Pre-Trend F-Tests

Outcome	F-statistic	p-value
French	3.51	0.015
Mathematics	1.15	0.326

Note: F-tests are joint tests of the significance of pre-treatment event-study coefficients (2017-2019), reference year 2021. Sample restricted to grade 6 cohorts entering 2017-2022.

A.5.2 Event Study Estimates

Table A.10: Event Study Estimates: REP+ vs REP

	French	Mathematics
	(1)	(2)
Treatment $\times$ 2017	−0.0295*** (0.0106)	−0.0081 (0.0106)
Treatment $\times$ 2018	0.0006 (0.0107)	−0.0161 (0.0109)
Treatment $\times$ 2019	−0.0067 (0.0102)	−0.0167* (0.0102)
Treatment $\times$ 2020	−0.0282*** (0.0092)	−0.0329*** (0.0092)
Treatment $\times$ 2022	−0.0133 (0.0092)	0.0049 (0.0092)
School FE	Yes	Yes
Year FE	Yes	Yes
Observations	658,631	658,600

Note: Dependent variable is standardized test score. Sample restricted to grade 6 cohorts entering 2017-2022, the only period during which REP remains untreated. Reference year is 2021. Standard errors clustered at the grade 5 school level in parentheses. ***p<0.01, **p<0.05, *p<0.1.

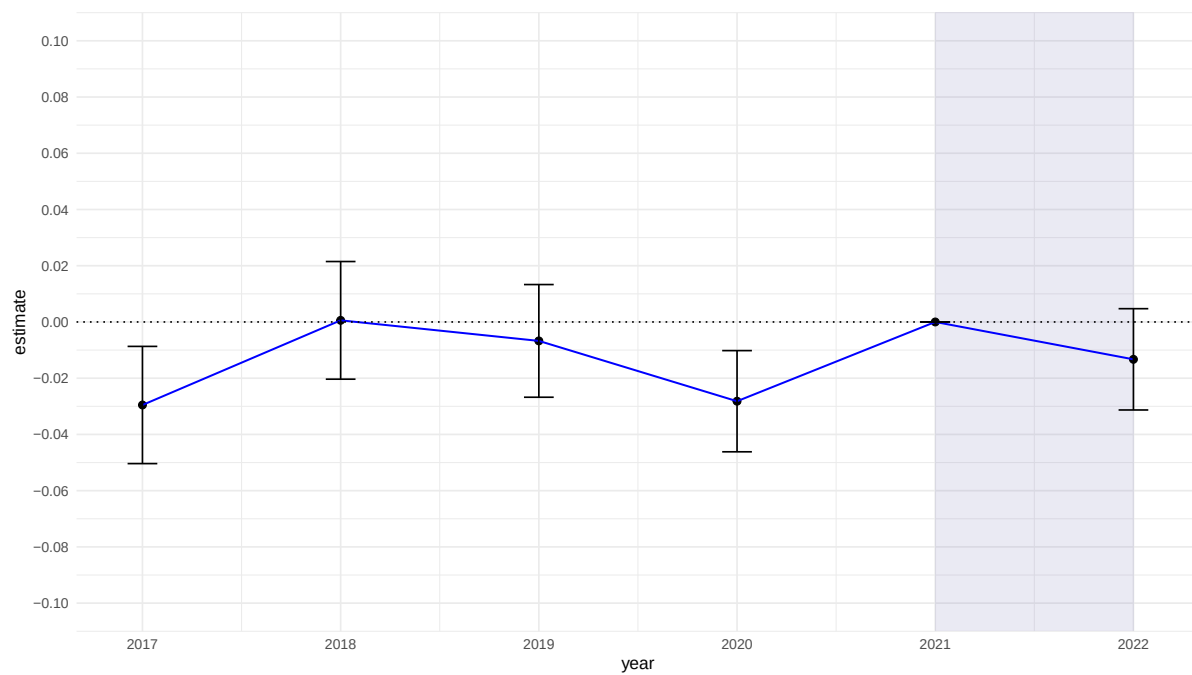


Figure A.5: Event study estimates: French score, REP₊ vs REP

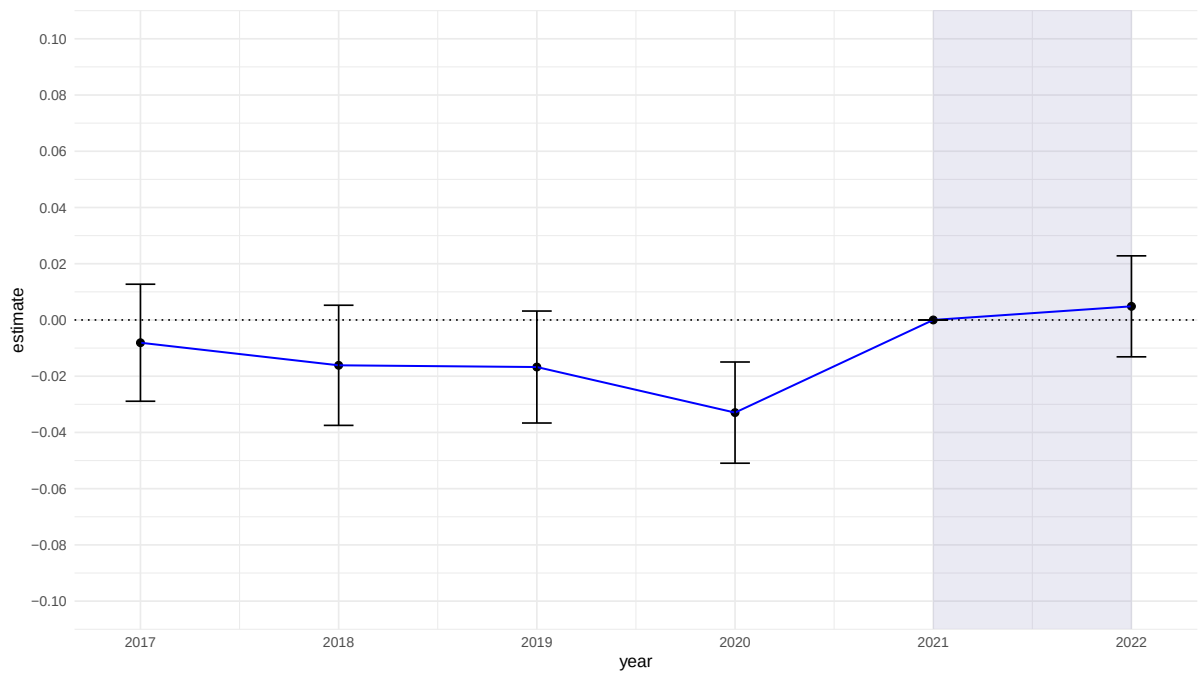


Figure A.6: Event study estimates: Math score, REP+ vs REP

A.5.3 Pooled Estimates

Table A.11: REP+ vs REP: Pooled Estimates

	French			Mathematics		
	Incl. 2020	Excl. 2020	COVID-ctrl.	Incl. 2020	Excl. 2020	COVID-ctrl.
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ Post	−0.0005 (0.0079)	−0.0046 (0.0082)	−0.0044 (0.0082)	0.0197** (0.0077)	0.0150* (0.0081)	0.0149* (0.0081)
Treatment $\times$ 2020			−0.0193* (0.0077)			−0.0229** (0.0077)
School FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	661,965	550,670	661,965	661,934	549,707	661,934

Note: Dependent variable is standardized test score. Sample restricted to grade 6 cohorts entering 2017-2022. Column (2)/(5) excludes 2020 from the sample; column (3)/(6) retains 2020 and controls for it via a Treatment $\times$ 2020 interaction. Standard errors clustered at the grade 5 school level in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Table A.12: REP+ vs REP: Including COVID Year (2020) in the Pre-Period

	French				Mathematics			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment $\times$ Post	0.0046 (0.0097)	−0.0004 (0.0079)	0.0045 (0.0097)	−0.0005 (0.0079)	0.0225* (0.0087)	0.0198* (0.0077)	0.0223* (0.0087)	0.0197* (0.0077)
School FE	No	Yes	No	Yes	No	Yes	No	Yes
Year FE	No	No	Yes	Yes	No	No	Yes	Yes
Observations	661,965	661,965	661,965	661,965	661,934	661,934	661,934	661,934

Note: Dependent variable is standardized test score (mean 0, SD 1). Sample restricted to grade 6 cohorts entering 2017-2022, the only period during which REP remains untreated. The pre-treatment average includes the 2020 COVID-19 disruption year. Standard errors clustered at the grade 5 school level in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Table A.13: REP+ vs REP: Excluding COVID Year (2020) from the Pre-Period

	French				Mathematics			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treatment $\times$ Post	0.0003 (0.0104)	-0.0045 (0.0082)	0.0003 (0.0104)	-0.0046 (0.0082)	0.0165* (0.0093)	0.0149* (0.0081)	0.0165* (0.0093)	0.0149* (0.0081)
School FE	No	Yes	No	Yes	No	Yes	No	Yes
Year FE	No	No	Yes	Yes	No	No	Yes	Yes
Observations	550,670	550,670	550,670	550,670	549,707	549,707	549,707	549,707

Note: Dependent variable is standardized test score (mean 0, SD 1). Sample restricted to grade 6 cohorts entering 2017-2022, the only period during which REP remains untreated. The 2020 COVID-19 disruption year is excluded from the pre-treatment average. Standard errors clustered at the grade 5 school level in parentheses. ***p<0.01, **p<0.05, *p<0.1.